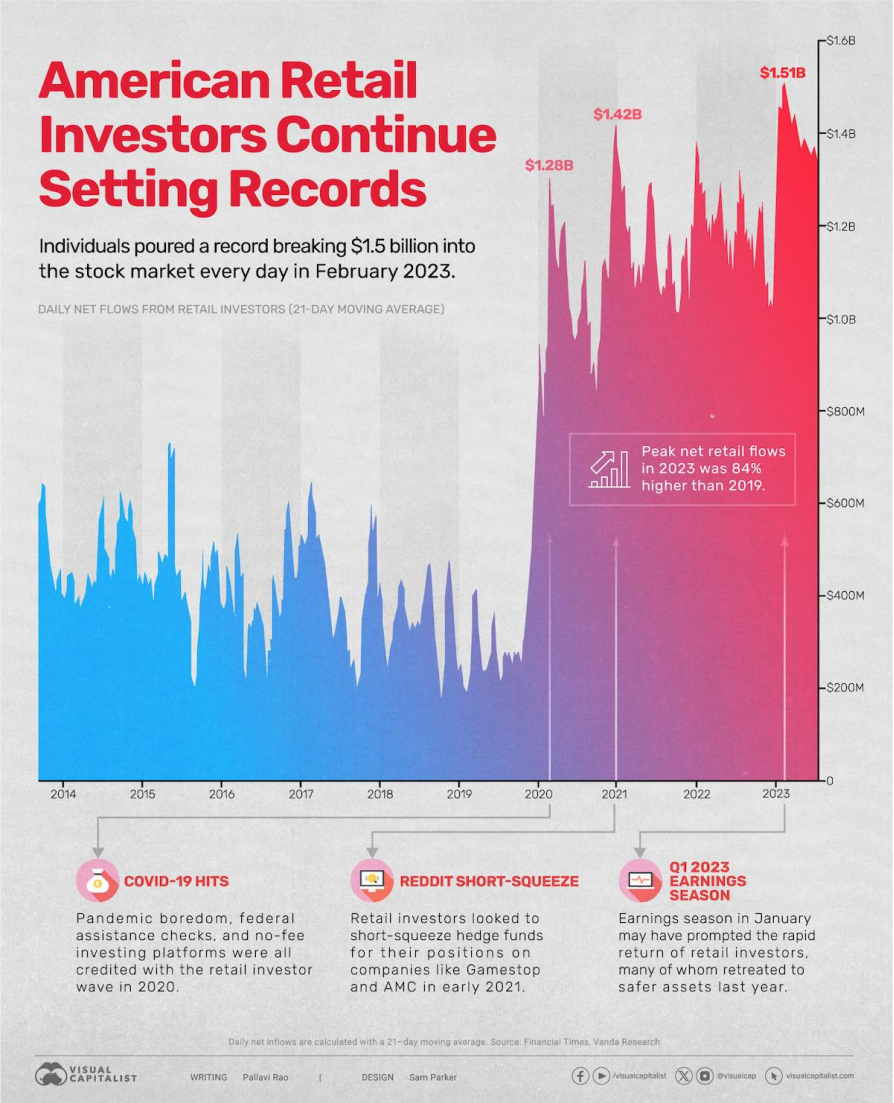
The Physics of Price Data

WHAT IS THE “MARKET” HOW DOES ONE VIEW IT FROM AN ALGORITHMIC PERSPECTIVE



The Retail Trading Revolution

- In 2020, retail traders accounted for ~10% of market volume
  - By 2025, this number has surged to nearly 25%
  - Large YoY increases motivated by the pandemic
  - Daily retail went from < 500 Million to 1.3 Billion post pandemic

The perfect storm

- The pandemic shut down the sportsbooks and casino's
  - Many turned to stocks an alternative to online gambling
  - Robinhood pioneered *zero commission trading*
  - The rise of Finance influencers

Who Are Retail Traders?

College students with stimulus checks

- 100 Million of the pandemic stimulus checks ended up in the stock market

Parents managing 529 plans

Retirees supplementing fixed income

Day traders chasing full-time income (spoiler: most fail)

YOU - learning to build systems

The Allure: Why People Trade

Trading is generally recognized as a more *sophisticated* form of gambling

- It is often confused with *investing*

The reasons someone might trade are multitude

- Financial independence fantasy
- Intellectual challenge
- Thrill
- Legitimate wealth building
- Sense of control in uncertain times
- Fear of Missing Out

What Makes Retail Trading Different?

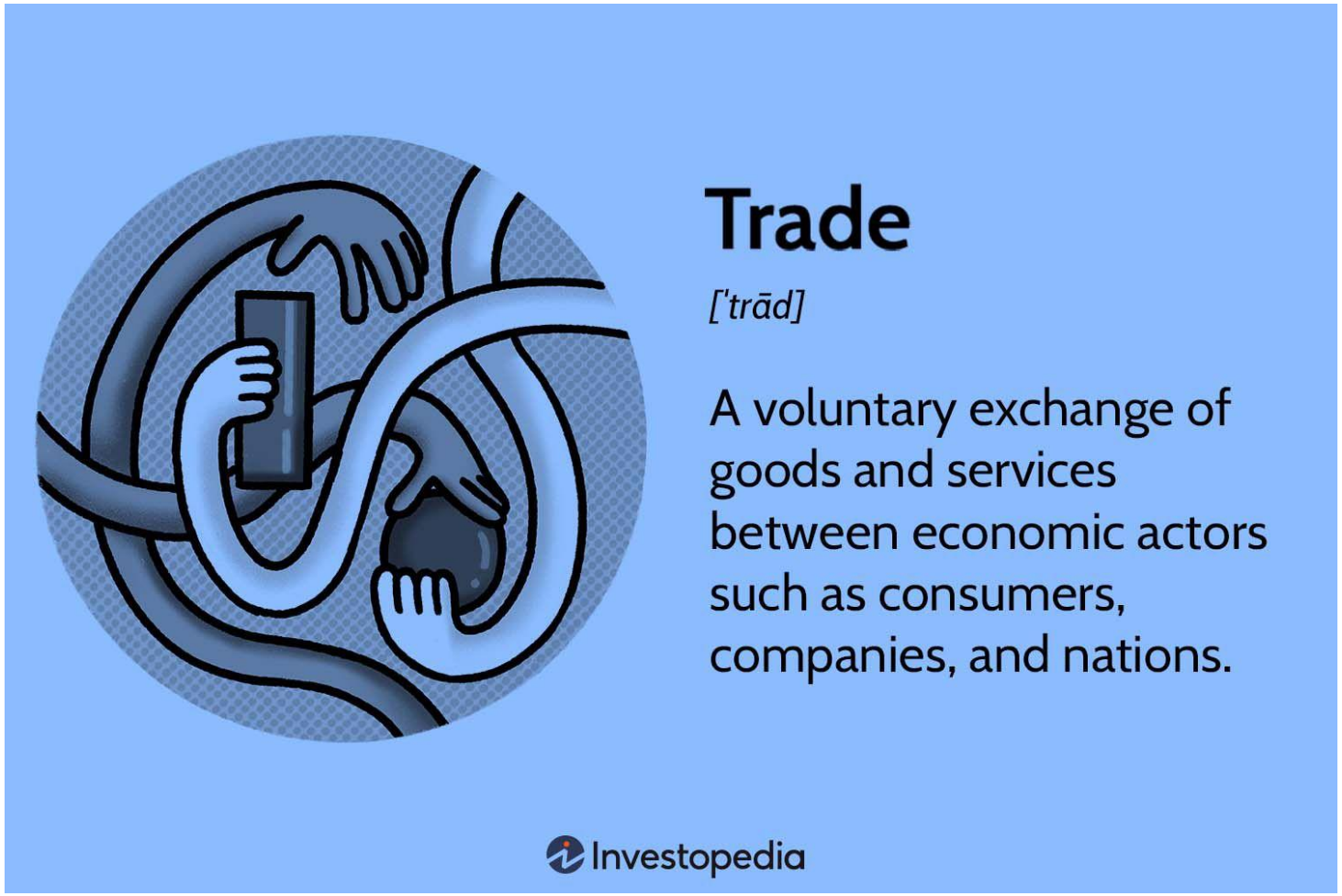
We are **not** Goldman Sachs with microsecond connections

A hedge fund with insider relationships

A market maker who sees the order book

We **are**

- Able to trade with delayed public information
- Competing against algorithms
- Learning to operate within constraints
- Working with seconds or more of delay



What Even Is Trading

Fundamentally

- You have money
- You think something will be worth more(or less) later
- You make a bet on that belief
- You were either right or wrong

What Is A **Trade**

A **trade** is an agreement on price, but a disagreement on value.

- In the financial world when a person buys something he/she thinks it will go up in price.
- The person selling the item thinks it will go down in value.

Whenever a person makes a trade, they decide to **enter** into the trade and **exit** out of the trade

- The **exit** is when you eventually *sell* the stock(hopefully for a profit)

As of ’25, the average daily share volume across all U.S. exchanges sits at around 12 to 18 billion shares

- NVDA alone often sees over 170 million shares traded in a single day
- 60 million individual trades a day
  - That same purchase of **5,000 shares** of Nvidia counts as only **1 trade**

This equates to about 10 trillion dollars worth of money changing hands globally each day

We typically see two main categories of trades: **Long** and **Short** trades

Typical Traditional Strategies

Retail traders typically fall into two broad categories based on how they make their buy and sell decisions

- Discretionary Trading
  - Decisions are subjective
    - May be based on advice, observations, intuition
  - Many challenges
    - Requires incredible *luck*
    - Fights against human emotions and stress
- System Trading
  - Uses mathematical models and approaches where decisions are predefined by system rules
  - Aims to remove guesswork and emotional bias

Trading Styles and Horizons

The "traditional" approach to the market is largely defined by the **holding period**, or how long a trader intends to stay in a position. As the timeframe shrinks, the reliance on rapid data increases.

- **Position Trading**
  - Months to years
- **Swing Trading**
  - Days to weeks
- **Day Trading**
  - Day only; no overnight positions
- **Scalp Trading**
  - Seconds to Minutes

Position Trading

Time Horizon
- Long-term

Philosophy
- Investors ignore short-term market "noise" and focus on the primary trend
- Position traders believe that identifying quality companies or assets and holding them through temporary volatility leads to superior long-term returns.

Goals
- Legitimate wealth building via market exposure through compounding returns over time

Key Characteristics
- Low Time Commitment
  - Periodic review(if that)
- Tax Efficiency
- Lower Stress
- Patience

Swing Trading

Horizon: Medium-term (Days to Weeks)

Philosophy:

- Swing traders aim to identify intermediate price movements that occur as markets oscillate between support and resistance levels.
  - Motivated by news catalysts

    Earning reports, product launches, regulatory decisions

Goal:

- To profit from price momentum while avoiding the intensity of minute-by-minute monitory, achieving monthly returns

Key Characteristics:

- Moderate Time Commitment: Requires daily chart review and position monitoring
- Risk/Reward Optimization: Typically targets 2:1 or 3:1 reward-to-risk ratios
- Strategy Mix: Combines technical and fundamental analysis

Day Trading

Horizon: Intraday (Hours; No overnight positions)

Philosophy:

- Eliminating "overnight risk" from major news that occurs while markets are closed. Day traders close all positions before market close to avoid gap risk from after-hours announcements.

Goal:

- To make money quickly
- Reality:
  - Many attempt this chasing a "financial independence fantasy," but it requires strict discipline to manage human emotions and statistical thinking to overcome natural cognitive biases.

Key Characteristics:

- Intense Time Commitment: Requires full attention during market hours (6.5+ hours daily)
- High Frequency: Can execute 10-100+ trades per day
- Rapid Decision Making: Entries and exits must be executed within seconds
- Emotional Challenges: Fear, greed, and frustration can derail performance

Why Day Trading

While the evolution of trading has moved from physical coat to digital securities, the underlying forces remain constant

The emotional challenges of *fear* and *greed* are actually just human expressions of shifting **supply** and **demand**

Intraday horizon(day trading), allows us to see the **law of demand** play out in seconds rather than years

Every trade you execute is a data point in the global search for **equilibrium**

What Drives *Price*

- Trading is the ultimate magnifying glass for they theory of supply and demand
  - First, when we say *supply* and *demand,* what do economists typically mean?

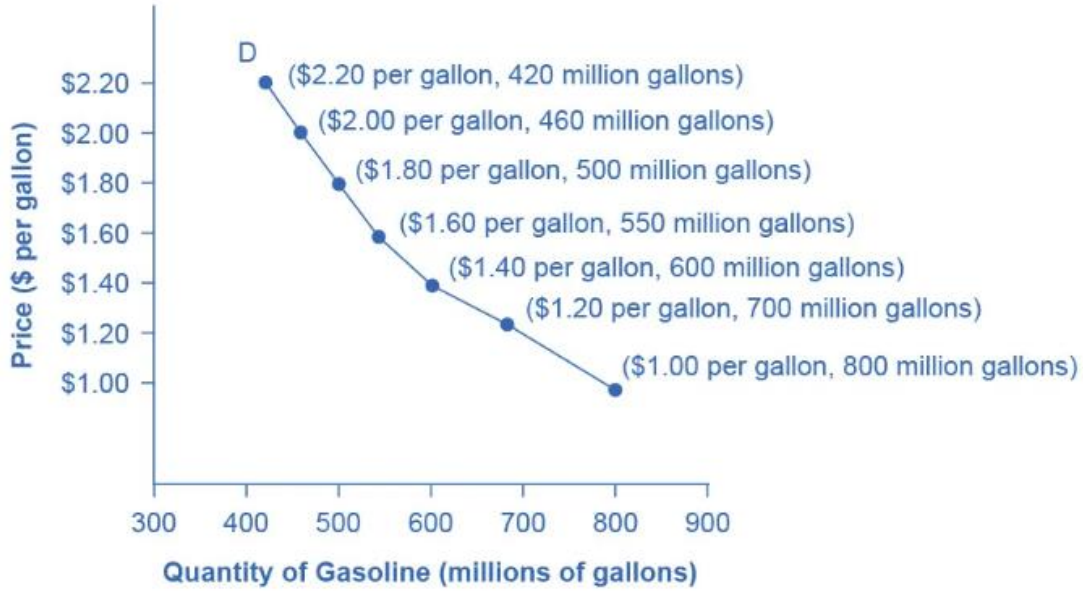
Economists use the term **demand** to refer to the amount of some good or service consumers are willing and *able* to purchase at given prices.

- Driven by the idea of needs and wants
- Demand is also driven by the ability to pay
  - If you cannot pay for it, you have no effective demand

The **price,** what the buyer pays, has a direct impact on the total number of units available for purchase at a given price, known as the **quantity demand**

- A rise in a security typically almost always decreases the quantity demanded
  - Pizza is more expensive - I buy less pizza
- A fall in the security increases the quantity demanded
  - Pizza is less expensive - I buy more pizza

The inverse relationship described above is **the law of demand**



Demand Curve

A demand curve is often used to visualize this inverse relationship

- Called a **demand schedule**

The visualization shows the price of gas

- Dollars per gallon of gas
- Quality demanded in millions of gallons over some time period

Supply of Goods and Services

When we talk about **supply** it refers to the amount of some good or service a produce is willing to supply at a given price

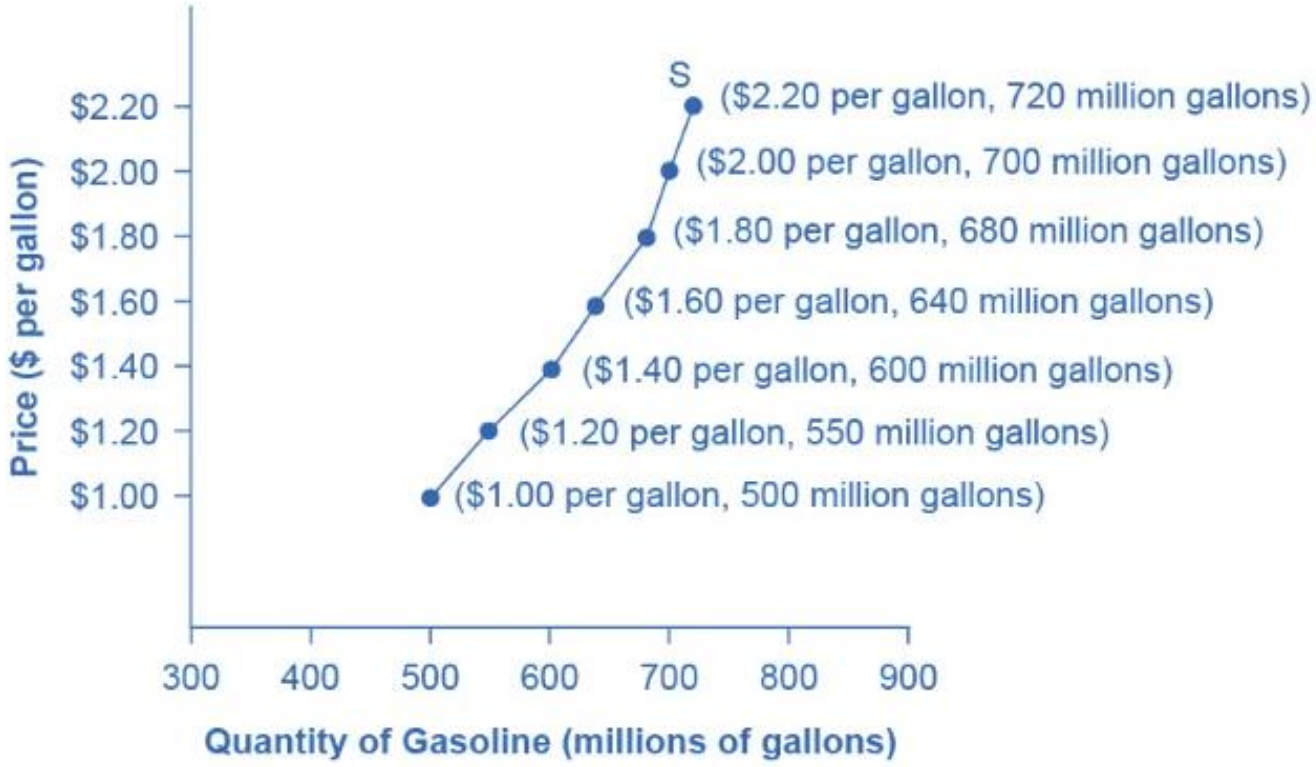
This relationship also exists

- A rise in price almost always leads to an increase in the **quantity supplied**
- A decline in price almost always leads to a **decrease** in the quantity supplied

Ex:

- When the price of gas rises, it encourages profit-seeking firms to try to expand oil reserves, drill for more oil, invest in pipelines…etc

This relationship is known as **the law of supply**

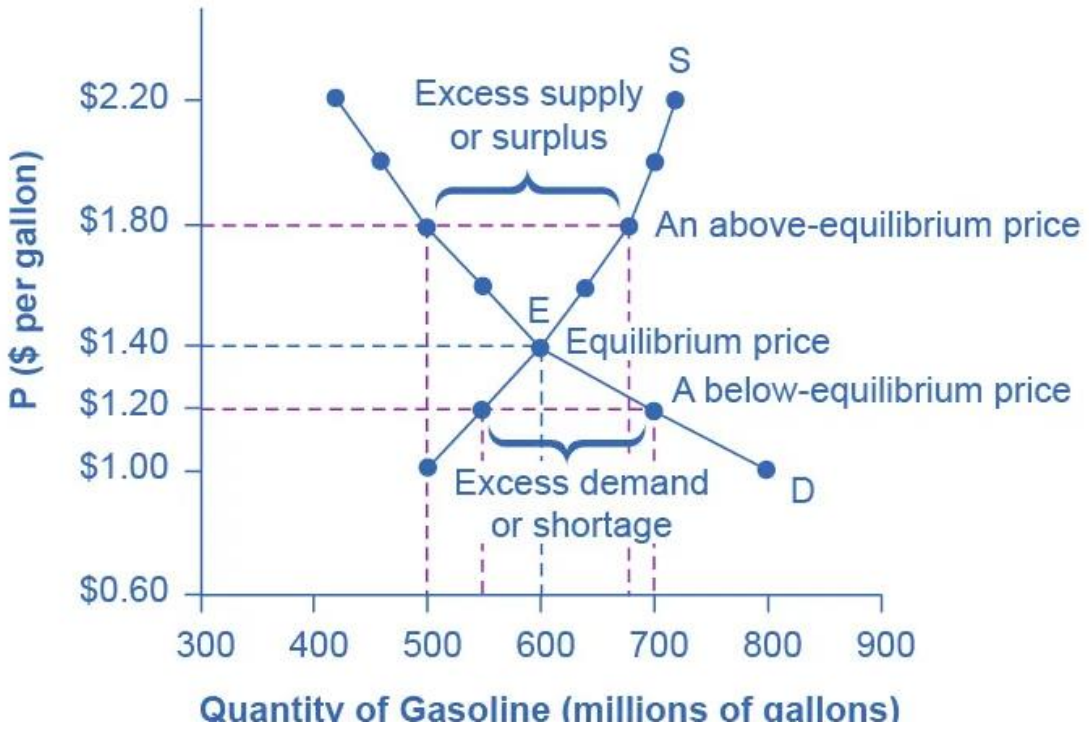


Supply Curve

We can once again graph this relationship using our gas example

A supply schedule shows the quantity supplied at a range of different prices

Where Demand and Supply Intersect



Together, demand and supply determine the price and the quantity that will be bought and sold in a market.

- The graph charts both of our gas examples
- When the lines intersect
  - E: equilibrium price
    - Where the amount of produce consumers want to buy equals the producer's supply

Q: Why don't we always sit at the equilibrium price?

- What happens when gas goes up?
- What happens when gas goes down?

Enter(again) "The Market"

We just saw that gas prices adjust when supply or demand shifts

- This process can take days, weeks, or years
  - Physical gas needs to be moved
  - Trade deals need to be worked
  - Pipelines need to be rebuilt

The market see's adjustments that happen in *milliseconds*

- The *ticker price* is just the current equilibrium(or close to it)
  - The exact point where some buyer demands met a seller's supply
  - $Q_D = Q_S$

Now, I keep saying *tickers,* but we have yet to really dive into this…so what is a ticker?

What are tickers?

Modern Stock Quote Features

Today's stock quotes include symbols, prices, and percent changes, providing investors with instant, crucial financial data.

Evolution from Ticker Tape

Ticker tape machines used analog signals for stock quotes, marking the start of real-time financial information flow.

- Invented in 1867 by Edward Calahan
- Worked as specialized telegraph receivers that printed stock symbols, prices, and volume onto a narrow paper strip.
- Before 1996, there was no real-time electronic ticker
- ticker tape was the standard, albeit delayed, source for market information
  - Phased out in the 60's with the rise of electronic signals

Words are hard…and slow

The machines were slow, initial models only printed about one character per second

Typing out a full company name like "The New York Central Railroad" would have caused massive data backlogs.

The technical problem

- We have **vast** amounts of technical information to convey to users with a finite numbers of characters and bandwidth
  - This is the **same fundamental technical challenge of speed/efficiency**
- We needed a way to abstract details away, could we abridge companies into “tickers” that convey the same information?
  - We do similar stuff all the time!
  - Morse code
  - Ascii characters, Unicode, sign language, French – we make up stuff all the time!

Stock *tickers* were a direct result of the above technical problem

The Birth of Stock Ticker

Using a 1 to 1 map

- Company names were converted to 1-4 letter abbreviations

This allowed exchanges to send 285 characters per minute in 1929

- Holy crap!

This change:

- Allowed active day trading possible for the first time as traders could now react to fluctuations as they happened
- It replaced word of mouth guesswork with an objective, printed record
- It allowed the NYSE to become the dominant exchange

It sounds perfect!

- Why bother changing!?!

Q: Why might we want to change this perfect system of ticker tape?

The Keg

Leading up to October 1929

- **Federal Reserve Rate Hike**
  - In August 1929, the Federal Reserve raised interested rates from 5% to 6%
- **The “Babson Break”**
  - Roger Babson, a famous financial forecaster predicted a terrific crash was coming
- **Hatry Crisis**
  - British investors pulled capital out of the US
- **Poor income distribution**
  - A small percentage of the population controlled a vast majority of the wealth
    - There weren’t enough new buyers to sustain the market once prices stopped rising
- **Tarriff Fears**
  - Debates over the smoot-Hawley Tariff Act(which raised import taxes) created a global trade war

This ultimately led to what is known as **black Thursday**

The Match(panic)

Tickers helped drive the market crash!

Oct 21 (Mon)

~6 million shares 100 minutes (1h 40m)

Ticker ran until 4:40 PM (market closed at 3:00)

Oct 24 (Black Thursday)

12.9 million shares

240 minutes (4 hours)

Oct 28 (Black Monday)

9.2 million shares

167.5 minutes (2h 47m)

Oct 29 (Black Tuesday)

16.4 million shares

152 minutes (2.5 hours)

The Result

Immediate Financial Losses

- The market lost
- One Day
  - $14 billion in value as over 16 million shares were traded.
- Two-Day
  - Between Black Monday and Black Tuesday, the market shed $30 billion in value
    - More than the total cost to the U.S. for World War I

Long-Term Economic Results

- Stocks continued to decline for nearly three years, eventually losing 89.2% of their peak value by July 8, 1932
  - The market did not fully recover to its 1929 highs until November 1954
- Approximately 7,000 banks, nearly a third of the U.S. banking system failed between 1930-1933
- Unemployment surged from 3.2% in 1929 to a peak of 25% by 1933
- Industrial production in the U.S. declined by nearly 50%, while the value of agricultural income fell by approximately the same amount.

Engineering Stability: Electronic Circuit Breakers

The 1929 Problem: Data Latency

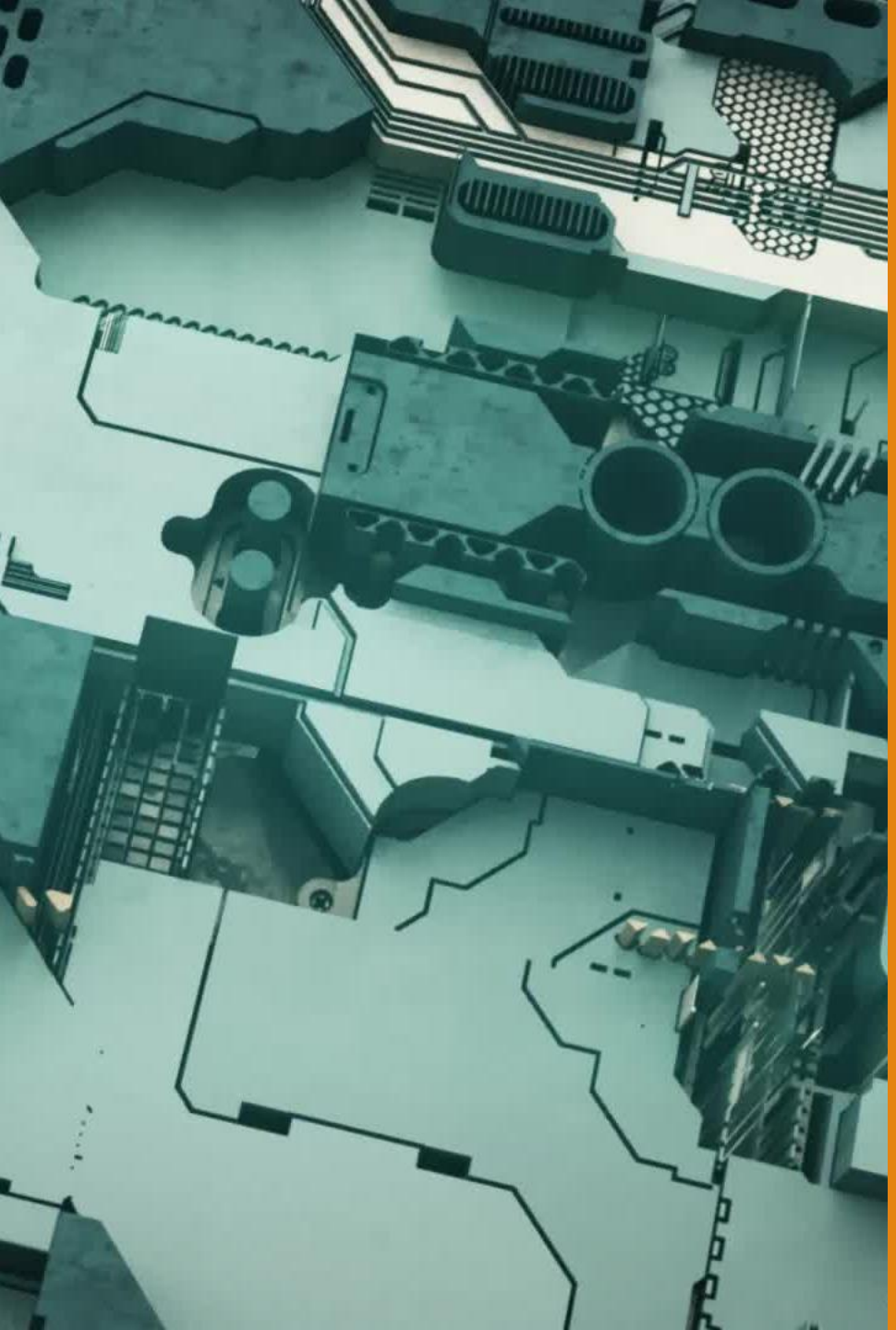
- The physical ticker tape ran **hours behind** real-time trading
- Investors sold in a *blind panic* because they didn't know the **truth price**
  - It's a car speeding up without a speedometer

Q: Is anyone into cars?

- What stops a driver from flooring a dodge demon to 170MPH?

The idea was simple

- Can we engineer systems to monitor the *velocity* of the market drop in milliseconds
- If we see things falling too fast, can we *trip* a breaker and force a pause?



From Tapes to Bytes

We already saw that 1971 saw the launch of the NASDAQ

- It was the worlds first electronic stock market
- Leveraged a network of computers to display quotes on screens rather than printing them on tape
  - **Old Tech:** Magnetic pulses turned a physical wheel to stamp ink on paper (max speed ~900 cpm)
  - **New Tech**: Digital packets sent via fiber optics and satellite

Legacy: By the mid-1970s, the "ticker" became a scrolling digital display (the "zipper") on building facades and television screens.

The *Address* of an Asset

The mapping problem

- Humans speak in names
  - Apple, Google, Ford
- Markets speak in Symbols
  - AAPL, GOOGL, F

Note:

- Tickers can be somewhat confusing
  - FORD is distinct from Ford Motor Company(f)

Our algorithm cannot trade Microsoft

- It can only trade MSFT

So, how do they know?

Where to Look?

There is no **central list** of all stock tickers

Tier 1: The Aggregators (Retail View)

- Google/Yahoo Finance
  - The standard for quick, human-readable lookups. Great for finding the ticker and seeing a "snapshot" of the price.

Tier 2: The Exchanges (Official View)

- NASDAQ / NYSE Listings: The official source of truth for what is listed.

Tier 3: The Data Providers (API View)

- Alpaca / Polygon.io: These provide "Asset Master" lists, downloadable JSON files of every tradeable asset available to your bot.
- There exist **many** possible data providers.

The Snapshot Illusion

When we look up NVDA on Yahoo Finance, you see a single number: $118.45

This is a **snapshot**

- It is static
- Likely delayed by 15 minutes
- Represents *one* specific transaction that happened in the past

The challenge of this semester is trying to trade as **close** to the *stream* as possible

- We want to get as close as we can to the data
- Graphs are *fantastic, more on them later,* but what happens when we stop looking at the snapshot and looking at the live data?

What do we Trade?

When we see NVDA at $183.45 what does this mean?

- Nvidia currently has 24.33 **billion** individual shares
  - 5% owned by insiders
  - 70% owned by institutional ownership
  - Rest is retail investors
- Each individual share is worth 183.45
  - $189.33 (Price) x 24.327 Billion (Shares) ≈ $4.619 Trillion

NVDA see's an approximate daily trade of about 181.7 **million** shares

- This is a **lot** of information, most of which retail investors **don't** typically need
  - It's important to understand as the raw data *drives* everything
    - Although we don't see it, it's the engine behind almost every decision matrix

From Charts to Events

Charts are vital to how we *see and understand* vast amounts of data quickly

- Yet, they abstract and hide a large amount of data from us

The **markets** themselves don't **move**

- Prices do not flow like water
  - They *teleport*
- A stock *price* is simply the *memory* of the last valid contract signed between two people
- Between 10:00:01 and 10:00:02, the price doesn't "travel"
  - it ceases to exist until the next event.

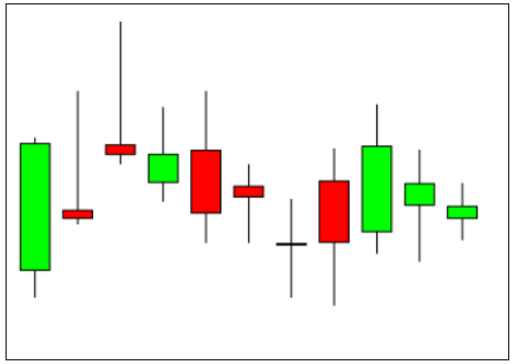
Markets are emitters

- Modern online markets broadcast discrete packets
  - This is one of the main reasons behind the NASDA's creation

The Data We *See* vs. The Data We *Get*

Any chart used to observe the market *lies* to you

- It is a *lie by omission*
- Charts hides **thousands** of trades inside a single green or red bar
  - Or graph!



The *ticker* is the **truth**

- Every row is a decision; every row is a fight between a buyer and seller

Most algorithms and trading decisions **fail**

- People are trained on understanding the chart, not the list

Time & Sales 3.4.73

Time	Trade	TSize	TExch	Bid	Ask	BSize	ASize	BExch	AExch	Date
10:10:33	38.30	200	NMS							20070710
10:10:33	38.30	100	NMS							20070710
10:10:33	38.30	800	NYSE							20070710
10:10:33				38.29	38.30	114	10	NMS	NYSE	20070710
10:10:33				38.29	38.30	114	9	NMS	NYSE	20070710
10:10:33	38.30	200	NYSE							20070710
10:10:33	38.30	300	NYSE							20070710
10:10:33	38.30	100	NYSE							20070710
10:10:33				38.29	38.30	114	8	NMS	NYSE	20070710
10:10:34				38.29	38.30	114	48	NMS	NYSE	20070710
10:10:34	38.30	100	NYSE							20070710
10:10:34	38.30	100	NYSE							20070710
10:10:34	38.30	100	NYSE							20070710
10:10:35				38.29	38.30	113	48	NMS	NYSE	20070710
10:10:35				38.29	38.30	113	47	NMS	NYSE	20070710
10:10:35				38.29	38.30	114	47	NMS	NYSE	20070710
10:10:36	38.30	200	THRD							20070710
10:10:36	38.30	200	NYSE							20070710
10:10:36				38.29	38.30	114	46	NMS	NYSE	20070710
10:10:37	38.30	100	NYSE							20070710
10:10:37	38.30	200	NMS							20070710

GE on 20070710

The Atomic Unit(the tick)

Time & Sales 3.4.73

Time	Trade	TSize	TExch	Bid	Ask	BSize	ASize	BExch	AExch	Date
10:10:33	38.30	200	NMS							20070710
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10:10:36				38.29	38.30	114	46	NMS	NYSE	20070710
10:10:37	38.30	100	NYSE							20070710
10:10:37	38.30	200	NMS							20070710

GE on 20070710

The following information is associated with a *trade*

- Timestamp | Symbol | Price | Size | Exchange
  - Stock or contract symbol, price, size(number of shares), Market Exchange(where the trade occurred) and a Timestamp

One trade is ultimately equivalent to one decision that has been made

An individual trade is referred to as a **tick**

Tick data is a stream of trades that may occur while the market is open

- Ex: a series of trades for GE:

During any particular day, there may be thousands if not **millions** of trades for a particular financial symbol

Tick Data is a Stream

Unlike our response from alpaca's API

- No notion of Open/Close

A tick is simply a discrete, successful transaction between a buyer and a seller

- It isn't just price, but volume and Venue

Time does **not** generate data

- If 100 shares are traded at $150, a tick is generated
- If nobody trades, *no tick is generated*
- *Bursts, silence, chaos*

What do we think happens for a company like NVDA?

The Stream Problem

NVDA generates millions of these ticks per day

- So do all high frequency stocks

Ticks do not happen every second

- Might have 0 ticks for ten seconds, 5,000 ticks the very next
- **The non-uniform problem**

We can’t possibly feed this amount of information into a naive system and expect to get good outcomes.

- Death by spreadsheet
- The problem is speed. During a crash, this list updates thousands of times a second.
  - Our moving average bot from last week cannot process 10,000 events a second, it would crash
    - So do pure software systems
- Just because we have speed, we don’t guarantee sophistication

Why we Aggregate(But Pay a Price)

Humans need summaries

- We cannot read a matrix scrolling at 5,000 lines per second. Charts are essentially "User Interface" for human eyes.

System Limit

- Processing every single tick requires High-Frequency Trading (HFT) infrastructure(microseconds)
  - N8N operates on seconds(or longer)

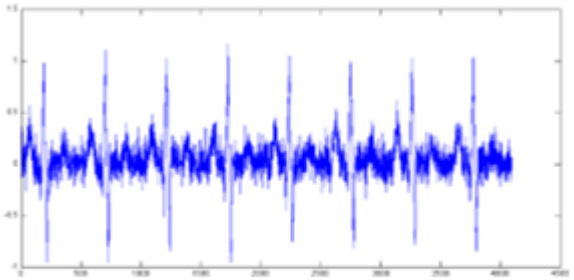
Our compromise

- Our goal is to aggregate
  - Trades truth for usability

Temporal data

We can use something known as **temporal data**
- This means that the data is time sensitive
  - Often ordered by time stamp

Time series data is a **construction,** not a direct observation
- The market operates on events, not minutes
  - We can try to enforce this idea of *time structure* as an artificial rhythm on a chaotic system



ECG Plot (Human heartbeat)

We typically see this represented as Time-Series data
- Collected on regular periodic basis
  - Every minute, every 5-minutes, every hour, every day, etc.

Open	High	Low	Close	Volume	Time	Date
4564.00	4565.25	4563.00	4564.75	3978	12:30	20230715
4565.00	4567.50	4564.00	4567.50	5082	12:35	20230715
4567.50	4567.75	4565.25	4566.00	7130	12:40	20230715
4565.75	4566.25	4564.00	4564.25	4080	12:45	20230715
4564.00	4564.75	4561.00	4564.75	5333	12:50	20230715
4564.75	4572.50	4564.50	4571.50	8543	12:55	20230715
4571.75	4572.25	4569.25	4570.50	8469	13:00	20230715

12:30 PM on July. 15th, 2023

12:35 PM on July. 15th, 2023

⋮

1:00 PM on July. 15th, 2023

We use the **sampling theorem**
- Our sample rate cannot be faster than our execution time
  - If n8N takes 2 minutes to run, trading a 1-minute bar is mathematically impossible.
    - We are reacting to Bar N while Bar N+2 is already forming
    - Perpetually *trading the past*

From Ticks to OHLCV

There is a trade at 8:14.59 AM at the price of 1266.50. The following tick data represents a set of trades from 8:15.01 to 8:20 AM:

This means that 2 persons (one buyer and one seller) exchanged money for a stock, contract, option, etc.

At 8:15.00,		Open is 1266.50
At 8:15.23	7	units are traded for **1266.25**
At 8:16.23,	10	units are traded for 1266.75
At 8:17.47,	2	units are traded for 1267
At 8:18.11,	10	units are traded for **1267.50**
At 8:19.34,	3	units are traded for 1267.25
At 8:19.51,	10	units are traded for **1267.00**

Low: Lowest traded during a given period.

High: Highest traded during a given period.

Close: Last traded price during a given period.

The **Volume** refers to the total number units traded during this period. A unit may be a share of stock, a contract, an option, etc.

- During the 5 minute period, 7+10+2+10+3+10 or 42 units were traded.

So during the 8:15.01 to 8:20 AM period the opening price was 1266.50, the highest traded price was 1267.50, the lowest traded price was 1266.25, and the last traded price was 1267.00. There were 42 units traded during this period.

Open	High	Low	Close	Vol	Time	Date
1266.50	1267.25	1266.25	1267.00	42	8:20	20010604

OHLCV Data Row

Open	High	Low	Close	Vol	Time	Date
1266.50	1267.25	1266.50	1267.00	42	8:20	20010604
1264.75	1265.25	1264.25	1264.25	54	8:25	20010604
1264.5	1266.5	1264.5	1265.25	785	8:30	20010604
1265.25	1267.75	1264.75	1266.5	2431	8:35	20010604
1266.5	1268.25	1266.5	1266.5	2549	8:40	20010604
1266.75	1267.25	1265.75	1266	2672	8:45	20010604
1266	1266.5	1263.25	1263.5	2872	8:50	20010604
1263.5	1264.75	1262	1262	2407	8:55	20010604
1262	1263.5	1261	1261	2415	9:00	20010604

In an **OHLCV** chart each row consists of 7 columns worth of data

- Open, High, Low, Close, volume, Time, Date

The **open, high, low, and close** describes the price movement during a 5-minute period

- The **Open** is the initial price and the beginning of the period
- The **Close** is the final price during the period
- The **High** and **Low** represent the highest/lowest traded price during the period

The **volume** represents how many shares (or contracts) were traded in the five minute interval

The **Date** attribute is a concatenation of Year, Month, and Day, or YYYYMMDD.

- So, **20010604** would be **06/04/2001**, or June 4th, 2001

The ***Time*** attribute is based on military time (00:00 to 23:55).

- This uses a 5-minute time frame, so it occurs in 5-minute intervals for this example.

OHLCV Granularity

OHLVC data *can* run at the five-minute interval, but there is no reason the same chart could not be generated differently over the same time series

5 – Minute Data						
Open	High	Low	Close	Volume	Time	Date
2265	2265	2263.5	2263.75	10777	9:05	20161214
2263.5	2263.75	2262	2262	14148	9:10	20161214
2262	2264	2261.75	2263.25	14875	9:15	20161214
2263	2263.25	2261.75	2261.75	15413	9:20	20161214
2262	2264.5	2261.5	2264.25	16187	9:25	20161214
2264	2265.25	2263.5	2264	19275	9:30	20161214
2264.5	2265.75	2263.75	2264	21355	9:35	20161214
2264	2264.75	2262.75	2263.75	17683	9:40	20161214
2263.75	2264.75	2263	2263	15289	9:45	20161214
2263	2263.5	2262.5	2263.5	11463	9:50	20161214
2263.25	2263.75	2262	2263.75	16489	9:55	20161214
2263.5	2264	2263	2263.5	11721	10:00	20161214

10 – Minute Data						
Open	High	Low	Close	Volume	Time	Date
2265	2265	2262	2262	24925	9:10	20161214
2262	2264	2261.75	2261.75	30288	9:20	20161214
2262	2265.25	2261.5	2264	35462	9:30	20161214
2264.5	2265.75	2262.75	2263.75	39038	9:40	20161214
2263.75	2264.75	2262.5	2263.5	26752	9:50	20161214
2263.25	2264	2262	2263.5	28210	10:00	20161214

30 – Minute Data						
Open	High	Low	Close	Volume	Time	Date
2265	2265.25	2261.5	2264	90675	9:30	20161214
2264.5	2265.75	2262	2263.5	94000	10:00	20161214

60 – Minute Data						
Open	High	Low	Close	Volume	Time	Date
2265	2265.75	2261.5	2263.5	184675	10:00	20161214

Pulling **OHLCV** Data

Let's think about this from a **workflow** perspective

- What might the general steps we want to **automate** be?
  - Well…how do I check an OHLCV chart normally?

Traditionally:

- We can simply lookup OHLCV chart for company X

Now do!

Create a workflow to:

- 1. Take a stock ticker from a user
- 2. ping an API and get the following
  - 5 minute OHCVL
  - 1 hour OHCVL
  - 1 day OHCVL
  - Another time frame of your choosing OHCVL
  - Graph the data

You may use:

- N8N
  - Only need to graph the close price
- Python
  - I have added a template on canvas for CS/AI/IT/FIN students who are feeling like expanding horizons
- Excel

Dropbox is open on canvas

- You must submit the artifact(excel file, n8n file, .py file)

What is *our* System?

Our system is **not** the market

Our system is **not**(yet) **the trader**

Our system is the **workflow** we design

Inside the system

- Data ingestion
- Feature computation
- Decision logic
- Execution attempt

Outside the system

- Market Participants
- Latency
- Competition
- Reality
- If its outside the system, we don’t control it, we try to design *around it*

Recap:

Why do we bother with OHLCV charts?

Why do we even bother with charts?

Charts are the **user interface of finance**

The human brain cannot process a scrolling list of 5,000 JSON object per second

- We are blind to the state of the market in its raw form

Charts are helpful to **verify** our logic

- **Not to execute our logic**
- Just like temporal data, we can project *geometry* onto *arithmetic*

Humans are pattern-recognition machines

- We look at charts to find patterns, then write code to detect it

Never mistake UI for the database

- **UI is lossy**
  - it renders a clean line where there was actual messy volatility

Graphing Financial Data

It is said that *a picture is worth 1000 words*

- In Finance, a chart can convey an extensive amount of information
  - Such as which direction the market is going along with the extent of a move

There are **many** ways that this data is visualized but two types usually dominate

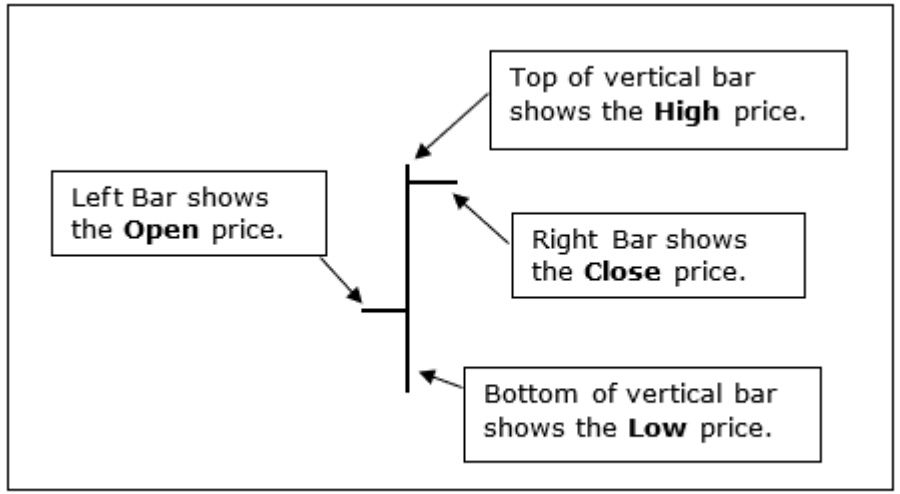
- Bar chart
- Candlestick charts

Both of these systems encode *human bias*

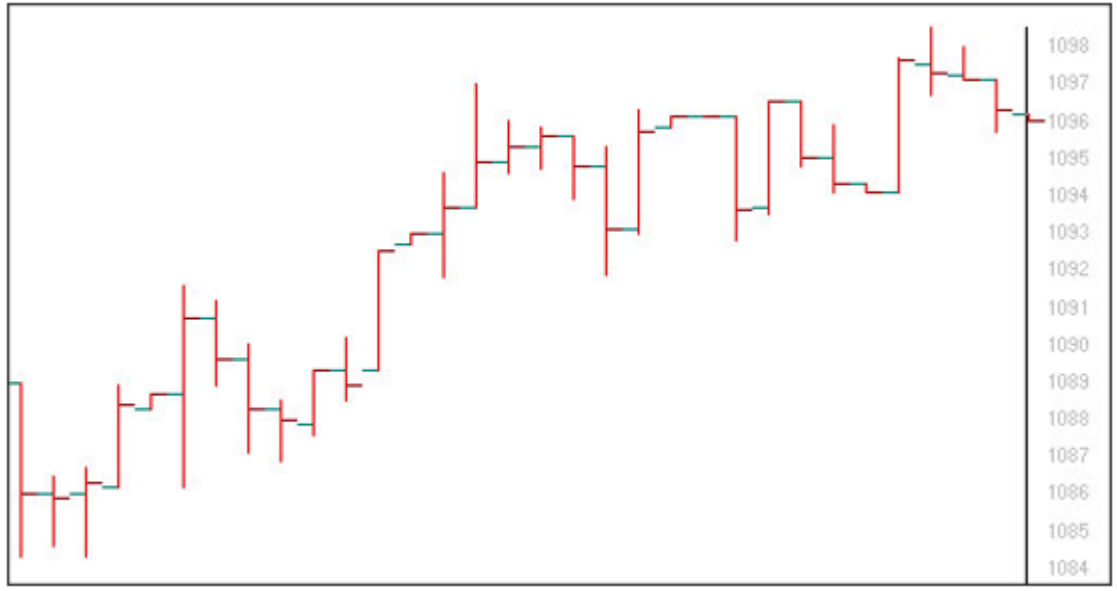
- Trends, momentum, pattern recognition
- These are things we *think* we can observe, but often it is bias, not truth

Bar Charts

The data found in a OHLCV chart is typically visualized as follows:



Components of a Bar in a Bar Graph



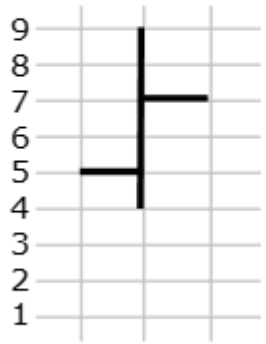
Sample Bar Graph (Chart) for a set of OHLC Values

Candlestick Chart

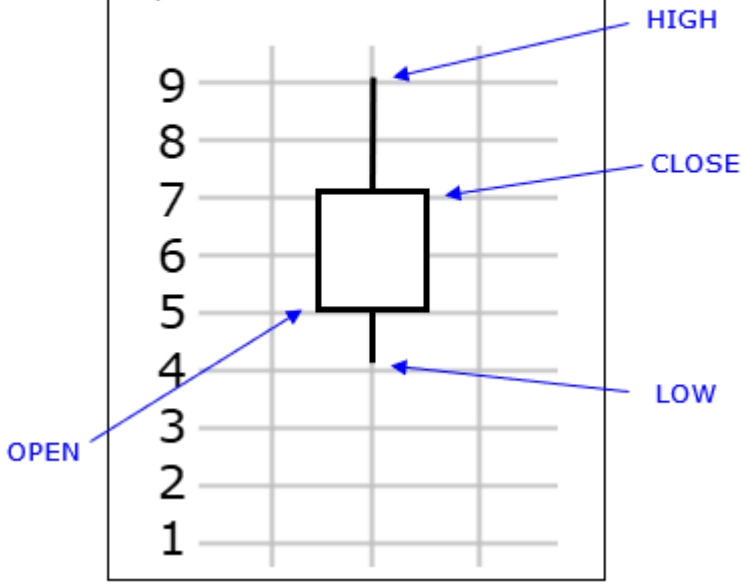
- A candlestick chart distinguishes between an up bar and a down bar
  - **Up:** CLOSE > OPEN
  - **Down:** CLOSE < OPEN

Suppose the OHLC is 5,9,4,7.

The bar graph would be represented as follows:

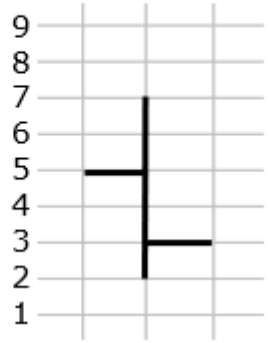


The candlestick would be represented as follows:

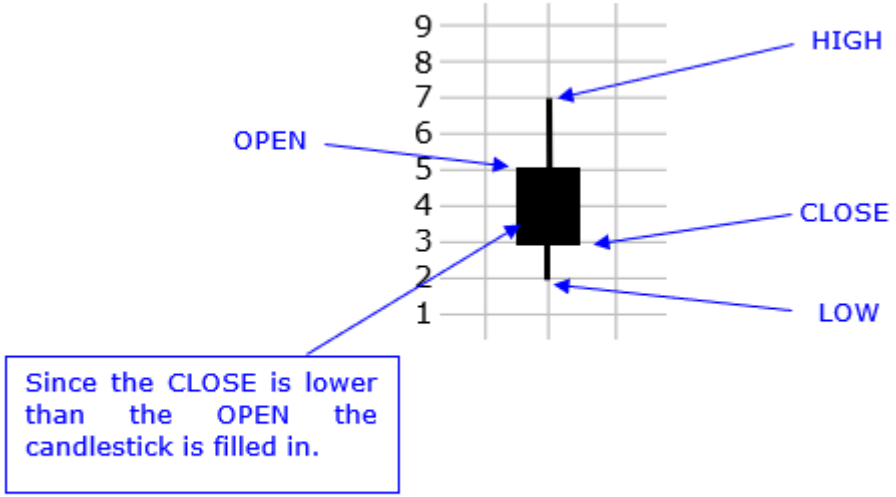


Suppose the OHLC is 5,7,2,3.

Bar graph:



Candlestick:

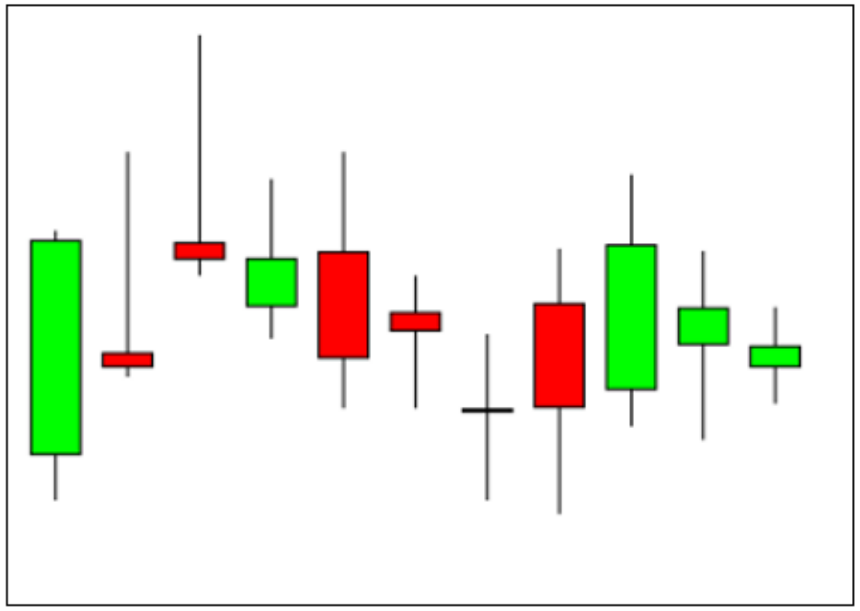


Candlestick Chart Visualized

Modern Web renders of candlestick charts help readers by using **green bodies** for up bars and **red bodies** for down bars

More and more traders default to the candlestick representation

- I personally think trading off them are fools' errands, but as a complement to other decisions, incredibly valuable.



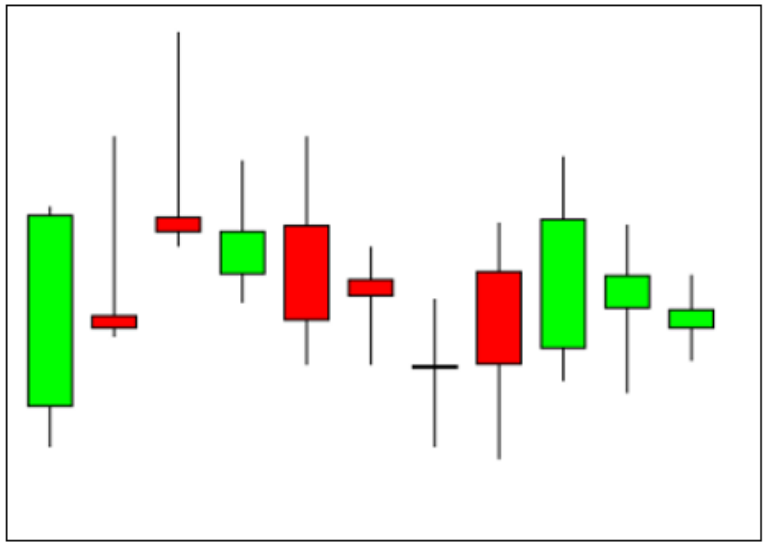
Charts are compression Algorithms

Time windows,

open/high/low/close

What do these candles hide?

- What does the data found in our raw json tell us that this candle doesn’t really?



Why systems fail when charts look right

How do machines learn?

How does this compare to the market?

Let’s Start Trading: Enter Trading

Long Trades

Normally, a person enters a trade by **buying** something, then exists the trade by **selling** it to someone else

When you **buy to enter** and **sell to exit** this process is called a **long trade**

A person who has entered a long trade, but has not exited the trade is considered to be in a **long position**.

- This is taken by those who believe a particular stock is going to **go up** in value
  - Ie: we buy AMD at $11 a share in 2017, sell it for $213 in 2025 and make about $202 per share

Short Trade

A short trade is when a person enters a trade by first **selling** something, then exits the trade by **buying** it back

When you **sell to enter** and **buy to exit** this is called a short trade

A person who has entered a short trade, but has not exited the trade is considered to be in a **short position**

- They are taking the stance that a particular stock is going to decrease in value

Say you **sell** a single share of NVDA stock at $220 and **buy** it back at $200, even though the stock itself went down 20 points, you still made a profit of $20

Going Short: How can you sell what you don't own?

A brokerage firm keeps track of all trades you make

In our **long trade** case, the firm would keep a record of how many shares(+N) you purchased, at what date, at what price

- When you exit, they would move your balance back to *zero* shares(-N)

In our **short trade** case, the firm would keep a record of how many shares you sold(-N)

- When you exit, they would move your balance back to *zero* shares(+N)
- You effectively *owe* the brokerage N firms that you buy at a later date

Orders are Commitments Under Uncertainty

Market Orders – “Fill it now”

- A request to buy or sell a security immediately at the best available current price
- **How it works**
  - You are a **price taker** you agree to whatever the current market participants are asking
    - We do this every time we shop at a store

Limit Orders – “Fill at my price”

- An order to buy or sell a security at a specific price or better
- **Buy limit**
  - Buy only if the price drops to $x or lower
- **Sell limit**
  - Sell only if the price rises to $y or higher

What are the risks of both?

Risks and Reasons

Market Orders

- **In volatile markets** the slippage might be high
  - Think back to our 1929 example earlier
    - People thought they were selling shares at $50, but due to time lag, they really sold it for $30

Limit orders

- You might *miss*

When should a trader use one over the other

- **Use market orders for:** highly liquid stocks with tight bid-ask spreads when you don't really care about a few cents of difference
- **Use limit orders for:** Low volume "penny stocks", after hours trading, or when you have a specific technical entry point based on research and indicators

Spread and Slippage

Spread

- There is a constant negotiation between buyers and sellers that creates a spread between the two sides
- The difference between the highest price a buyer will pay and the lowest price a seller will accept for an asset
  - If you want to make an instant purchase, you need to accept the lowest ask price from a seller.
  - If you'd like to make an instant sale, you'll take the highest bid price from a buyer.

Slippage

- The gap between the price you expected to trade at and the price your order actually fills at
- This is a dynamic cost that arises when your order is filled
  - Especially common when dealing with rapidly changing prices
- Let's pretend you want to buy 10 shares of a stock, and the only other orders in the market are limit orders. Current accepted (bid) price is $4.99, and there are 5 limit sell orders (ask) at $5.00, 3 at $5.01, and 2 at $5.02
  - You submit a market buy order, expecting to pay $4.99. But because there were no shares on offer for that price, your order bought the shares at the next best prices: 5@$5.00, 3@$5.01, and 2@$5.02 for an average cost of $5.007. Your order 'slipped' from $4.99 to $5.02.

The Bid-Ask Spread

The “Bid” and the “Ask” represents the two sides of a negotiation happening in real-time

The bid: The highest price a **buyer** is willing to pay

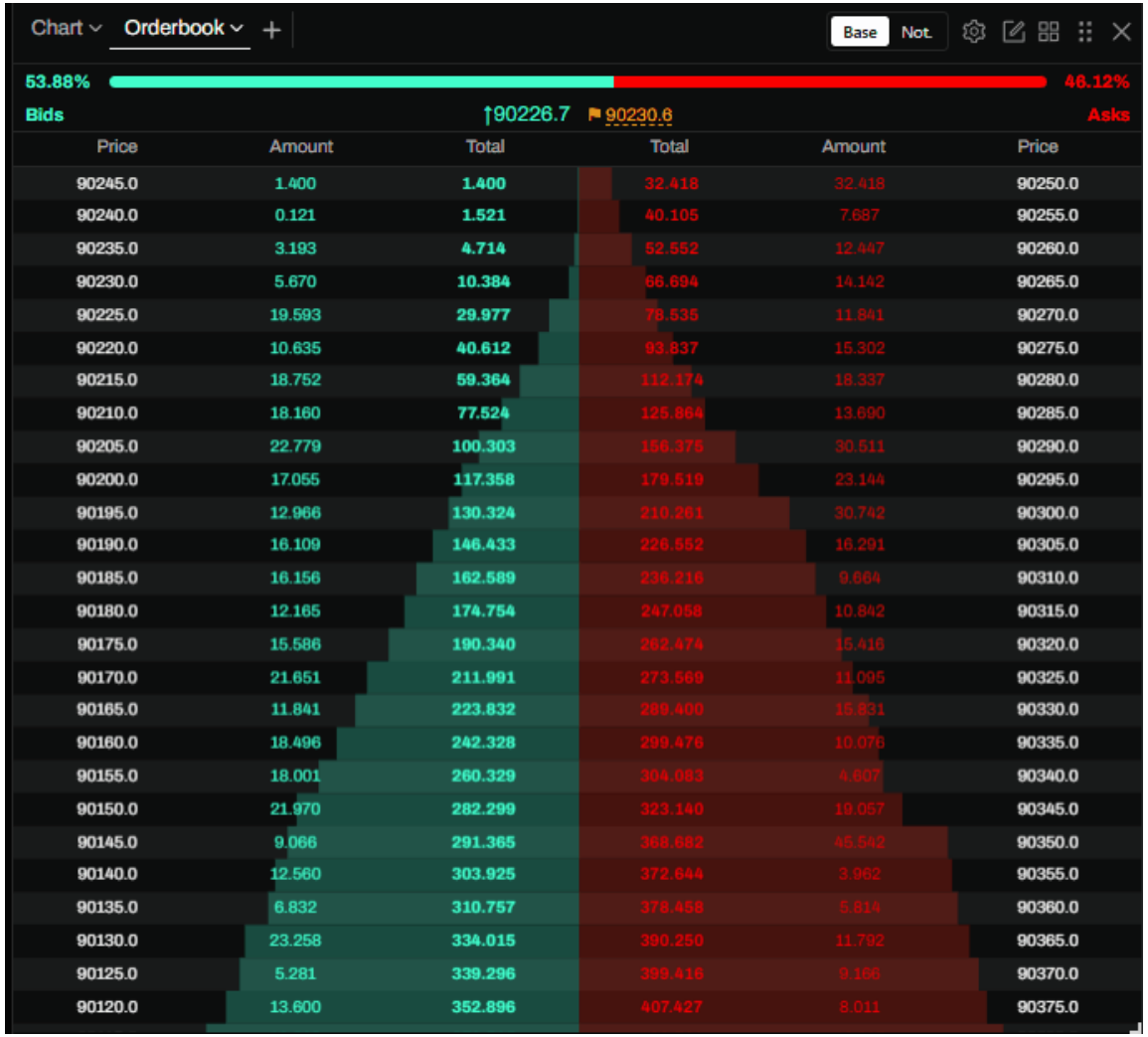
- Demand

The ask: The lowest price a seller is willing to accept

- Supply

Our **Spread** is the difference between these two prices(Ask-Bid)

- This is exactly how our **market makers** earn their profit by providing liquidity to the exchange
  - A **narrow** spread ($0.01) means the stock is highly liquid
  - A **wide** spread ($0.50) means the stock is thinly traded and expensive to enter



Price	Amount	Total	Total	Amount	Price
90245.0	1.400	1.400	32.418	32.418	90250.0
90240.0	0.121	1.521	40.105	7.687	90255.0
90235.0	3.193	4.714	52.552	12.447	90260.0
90230.0	5.670	10.384	66.694	14.142	90265.0
90225.0	19.593	29.977	78.535	11.841	90270.0
90220.0	10.635	40.612	93.837	15.302	90275.0
90215.0	18.752	59.364	112.174	18.337	90280.0
90210.0	18.160	77.524	125.864	13.690	90285.0
90205.0	22.779	100.303	156.375	30.511	90290.0
90200.0	17.055	117.358	179.519	23.144	90295.0
90195.0	12.966	130.324	210.261	30.742	90300.0
90190.0	16.109	146.433	226.552	16.291	90305.0
90185.0	16.156	162.589	236.216	9.664	90310.0
90180.0	12.165	174.754	247.058	10.842	90315.0
90175.0	15.586	190.340	262.474	15.416	90320.0
90170.0	21.651	211.991	273.569	11.095	90325.0
90165.0	11.841	223.832	289.400	15.831	90330.0
90160.0	18.496	242.328	299.476	10.076	90335.0
90155.0	18.001	260.329	304.083	4.607	90340.0
90150.0	21.970	282.299	323.140	19.057	90345.0
90145.0	9.066	291.365	368.682	45.542	90350.0
90140.0	12.560	303.925	372.644	3.962	90355.0
90135.0	6.832	310.757	378.458	5.814	90360.0
90130.0	23.258	334.015	390.250	11.792	90365.0
90125.0	5.281	339.296	399.416	9.166	90370.0
90120.0	13.600	352.896	407.427	8.011	90375.0

Order Book Example

This is an example of an **order book**

- Its essentially a live scoreboard of every buyer and seller currently waiting *in line* to trade

The two camps

- **Bids(Green lines)**: These are the **buyers.** They are submitting limit orders saying, “I want to buy, but only if the price drops to X”
- **Asks(red lines):** These are the **sellers**. They are submitting limit orders saying, “I want to sell, but only at price Y or higher

The current price and spread

- **The best bid:** The highest price someone is willing to pay right now is 90,245
- **The best ask:** The lowest price someone is willing to sell for right now is 90,250
- **The spread**: There is a $5 gap, no trade happens until a *market order* comes through

How do investors make buy and sell decisions?

This is perhaps the **golden** question of what makes a *trader*

There are many ways that people make buying and selling decisions

- What to buy(or sell)
- When to buy(or sell)

It is possible, and more common, to weave these two decisions together into a single model as you analyze buy and sell signals

- But first, let's consider each method

What to buy(or sell)

Traders typically consider the financial health of a company in order to determine whether to trade stock in that company

- This may involve
  - Examining price to earnings(PE ratio)
    - Are profits increasing year over year
    - Is the company actually selling more over time
  - Examine the Debt-to-Equity Ratio
    - How much debt does the company currently hold
  - Free Cash Flow
    - How much cash the company actually generates after expenses
  - Management Quality Indicators
  - Industry and Macro Factors
    - Market share trends
    - Sector strength
    - Regular

When to buy(or sell)

This is the practice of **market timing**

- The use of charts or signals to identify patterns
- This type of trader might try to identify the top or bottom for a particular stock

This trader can consider multiple angles

- Fear/greed in the sentiment of the market
- Overall wider economic news and macro announcements
- Volume spikes
  - Confirm buy/sell pressure
    - Fomo in the markets

Buy the rumor, sell the news

Extra Slides

IN NO PARTICULAR ORDER OF RELEVANCE

How they fit together

These two methods of trade(what/when) are often combined into two general categories of traders

Discretionary

- In discretionary trading buy/sell decisions tend to be subjective
- Decisions might be made based on advice from a perceived expert, observations, or even intuition
  - The hardest challenge of discretionary trades is managing human emotions
    - The pain of watching your asset fall is a real stressor

System Trading

- System trading uses a mathematical approach to making trades
- Most buy/sell decisions are predefined by rules of the system
  - This takes the guesswork out of placing trades and ideally allows you to place trades with a high chance of success
    - Does not guarantee profits

Algorithmic Trading

Trade Execution

- Dealers generally execute their orders through a shared centralized order book that lists the buy and sell orders for a specific security ranked by price and order arrival time
  - FIFO
- This centralized order-driven trading system continuously attempts to match buy and sell orders

The **Order Book**

- A system divided into two parts
  - Buy orders – highest price at the top
  - Sell orders – lowest price at the top
- Orders are generally listed by price and time priority, which means most exchanges prioritize orders based on the best price and FIFO framework

Order Book – ABC Inc.

Buy		Sell	
Quantity	Price	Price	Quantity
5,000	99	99	4,000
8,000	98	100	10,000
10,000	97	101	1,000
15,000	95	103	15,000

(a) Before matching a trade

Order Book – ABC Inc.

Buy		Sell	
Quantity	Price	Price	Quantity
1,000	99	100	10,000
8,000	98	101	1,000
10,000	97	103	15,000
15,000	95	104	3,000

(b) After matching a trade

What is Financial Data Mining

Financial Data Mining is the process of identifying useful mathematical or graphical-based patterns in order to make effective buy and sell decisions

- As of ‘24 and ‘25 Forbes describes data science as the **heart** of modern algorithmic trading
  - The most successful firms now rely on proprietary Machine Learning and Deep Learning models to mine market sentiment and perform risk modeling
- It’s estimated that 60% to 80% of the U.S. trading Volume is done by algorithmic decision making

Models attempt to predict and identify patterns within financial data sets, with the hopes of exploiting them

- What is a Data Set in the context of the stock market?

Timing the Market

Day traders attempt(and fail) at trying to time these up and down swings

- Yet it is possible to take full advantage of both upward and downward price movements
- In a perfect world, you would go long at the green arrows, go short at the red arrows

NVDA Stock from 1999 to 2019

System Trading

System Trading can be **semi-automated** or **automated**

A **semi-automated** trading method typically involves using discretionary methods to enter trades, while relying on automated signals to exit positions

- A trader may enter a long position if the price goes above a trend line, once the trade is entered, the traders computer will use fixed prices to exit the trade
  - The inverse is true, this is a exit position vs stop loss(loss limit)

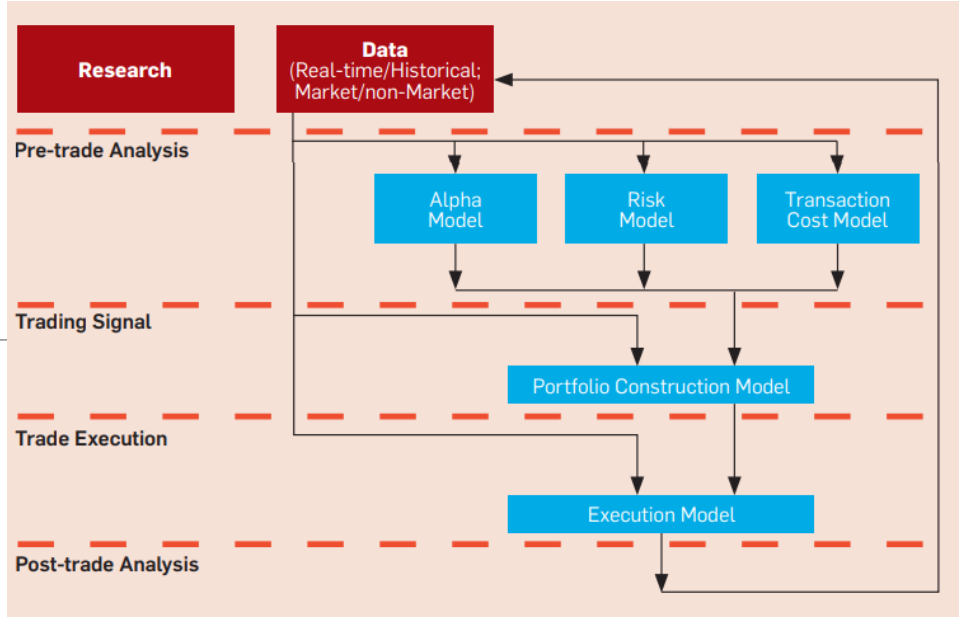
Automated trading occurs when the traders platform performs all trade executions based on a customized trading strategy or a commercially available system to develop a custom strategy

- This is typically a **lot** of programming and complexity
  - Commercial platforms exist such as TradeStation's EasyLanguage, C#, Java, or the most modern popular method **of Python**

Our class is going to focus on both **when and what** to buy and **system trading**

- Our end goal is to go from **semi** to a completely automated trading bot
  - To do this, we will explore *technical analysis*(more on this later)

System Trading



Generally, a system trading approach involves five general phases

Data Access/Cleaning

- Obtaining and cleaning data that drives the system

Pre-Trade Analysis

- Analyses properties of assets to identify trading opportunities using market data or financial news

Trading Signal Generation

- Identifies the portfolio of assets to be accumulated based on the pre-trade analysis

Trade Execution

- Executing orders for the selected asset

Post-Trade Analysis

- Analyzes the results of the trading activity, such as the difference between the price when a buy/sell decision was made and the final execution price
- Most systems in industry are devoted to the data access/cleaning and pre-trained analysis stages
  - Latter stages of trade are managed by humans(or closely supervised)

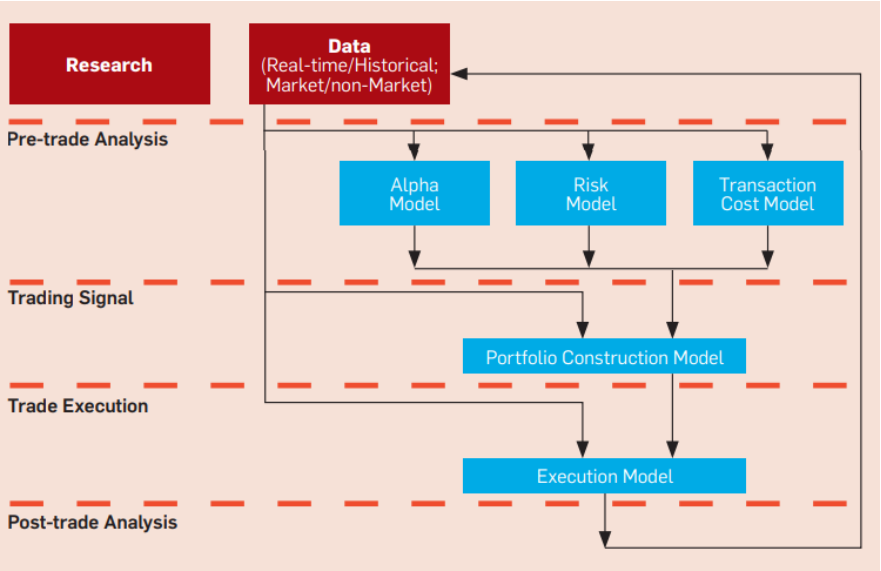
Pre-Trade Analysis

The pre-trade analysis phase

- Analyses properties of assets to identify trading opportunities using market data or financial news

This often comprises computing analytical features through three main compoenets

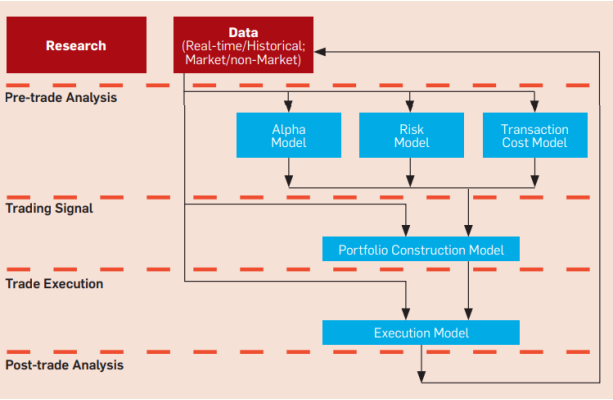
- **Alpha Model**
  - The mathematical model designed to predict the future behavior of the financial instriments that the algorithmic system is intended to trade
- **Risk Model**
  - This evaluates the levels of exposure/risk associated with the financial instruments
- **Transaction cost model**
  - Calculates the potential costs associated with trading the financial instruments



Trading Signal Generation

Trading Signal Generation

- Identifies the portfolio of assets to be accumulated based on the pre-trade analysis



The portfolio construction model takes as its inputs, the results of

- Alpha model
- Risk model
- Transaction cost model

It then decides what portfolio of financial instruments should be owned and in what quantities in the next time horizon

- It’s the pre-decision phase

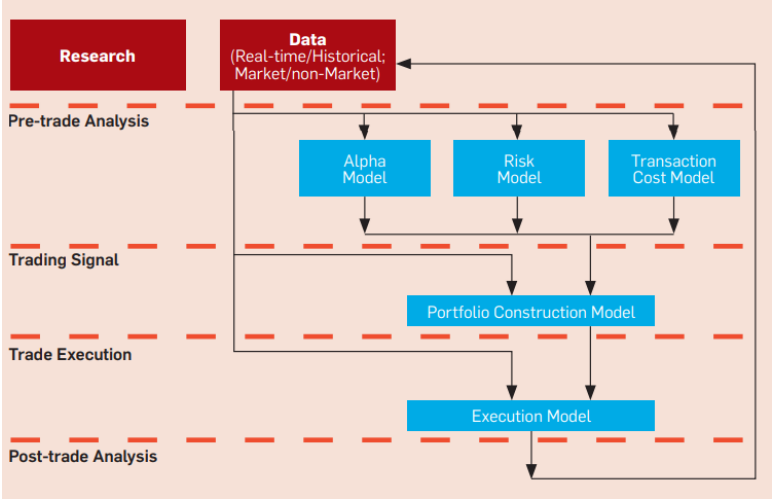
Trade Execution

Trade Execution

- Executing orders for the selected asset

After the trading signal has been generated and the portfolio constructed, the *execution model* performs several decisions with constraints on the actual transaction costs and trading duration

- The most general decision is the trading strategy followed by the venue and order type
  - Decisions are handled by smart order routing



Post-Trade Analysis

Post-Trade Analysis

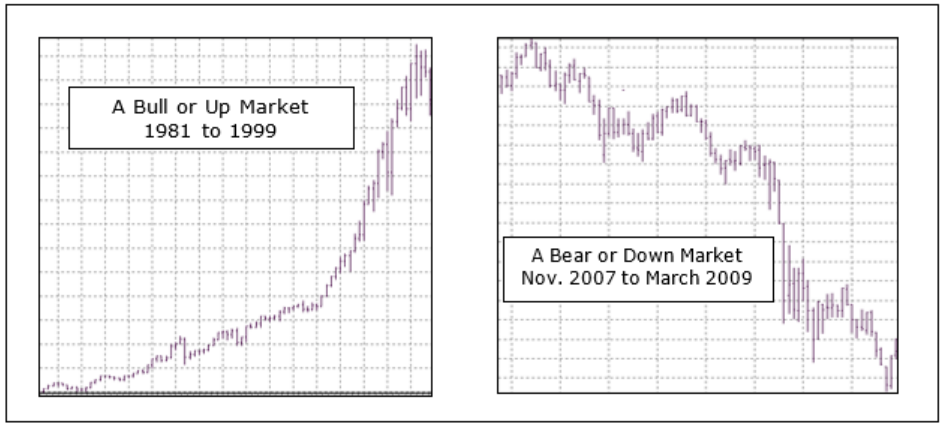
- Analyzes the results of the trading activity, such as the difference between the price when a buy/sell decision was made and the final execution price
- A mathematical evaluation between the expected price and the final execution price

This difference between expected and final execution is **slippage**

Bulls and Bears

One of the **first** things to determine is market direction

- **The trend is your friend**
  - If you can determine market direction, then it is much easier to succeed
    - Simply go with the flow
- The following graphs show an evident direction, which is called a **trending market**



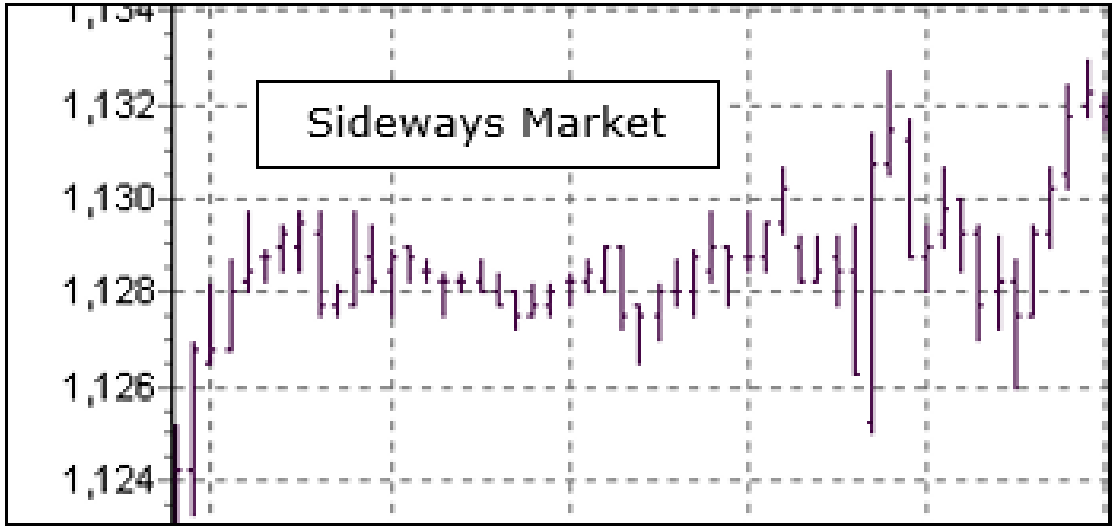
Trending type of markets: The market has a definite direction.

Bulls and Bears continued

Depending on the timescale, it can be harder to visualize and monitor trends in the market

Indeed, sometime the market does **not** have a clear direction

- In this instance, we call this a **sideways market**



Financial Instruments

Up to this point, we have been talking about *trades* and *equities,* but we have not really addressed what a **stock** is.

A **stock** is a security issued in the form of shares that represent ownership interests in a company. A person trades stock by buying and selling **shares**.

- Companies that are **publicly traded** allow retail investors to purchase shares
  - Google, Nvidia, Meta, IBM, Intel, Southern Company, AT&T, so on

The U.S. economy has become progressively more financialized

- This involves extending what is possible to *trade* far beyond traditional markets.
- We are living in an era where nearly everything, from memes to predictions, is becoming a tradable asset
  - This is bad. This is unarguably bad. Do **not** engage in this
    - The stock market is generally gambling with *some regulations*
    - this is degenerate gambling with no regulation

From Activities to Markets: Entertainment

Entertainment

- The sheer scale of capital moving through sports betting apps has shifted from a niche activity to a massive macroeconomic force in *six years*
  - In 2018, Americans legally wagered roughly **6.6 billion**
  - In 2025, Americans legally wagered roughly **150 billion** a 2,170% increase in volume
- This is further compounded by the rise of *micro-betting*
  - The fastest-growing segment(15% a year) allows users to be on individual plays in real time
    - Ex: will the next pitch be a strike

Betting discussions now make up a significant portion of sports broadcasts

- Operators spend over 1.2 **billion** annually on marketing
- Entire *industries* are propped up **just** to mainly serve as an outlet for gambling
  - esports

From Activities to Markets: Tokens

Tokens: This *was* previously the most aggressive form of financialization(more on its successor in a moment)

- Tokens are a way to turn literally any digital idea, community, or joke into a liquid, tradable asset in seconds.
- It is the commercialization of *internet protocols*
  - Blockchain technology

As of 2025, there are over 37 million unique tokens in existence

- In 2021, there were fewer than 100,000

Platforms allow users to launch a tradeable asset for less than $2 in fee's, often reaching million-dollar market caps within hours

- To be quickly dumped(scams)

Approximately 28% of Americans now own cryptocurrency as of 2025

- 31% of investors hold *memecoins* such as DogeCoin
  - This literally turns cultural jokes as core portfolio positions

From Activities to Markets: Memecoins

Memecoins are the purest example of where the *value* is **not** utility but the strength and attention of an online community

Memecoins soared to over **$140 billion** in 2025

- 5% to 7% of crypto industry value

Attention based volatility

- Cultural moments are now indexed in real time
  - During the ‘24 election cycle $TRUMP reached market caps of **$30 billion** allowing people to *go long* or *go short* on political momentum rather than just voting or donating

While millions of tokens are created, only 1.4% reach significant liquidity

- These communities are fleeting, speculative bubbles

From Activities to Markets: Prediction Markets

Prediction markets represent the final stage of the financialization of the US

- **Beliefs themselves** are converted into liquid asset classes
- Strip away the token, you invest directly on the outcome
  - Replace an **opinion** with a **position**

The financialization of Truth

- The scale of prediction markets has moved from academic curiosity to a multi-billion dollar information infrastructure
- Polymarket and Kalshi reached a record **$10 billion** in a **single month**
- Prediction markets are often seen as the source of truth, moving hours before traditional news networks and polls

24/7 prediction cycle

- Macroeconomics
  - Wil the fed cut rates
- Pop culture
  - Will a specific celebrity couple break up
- Science/Tech
  - Will SpaceX land starship on the first try

Modern Financial Data Mining

No **longer** just raw stock performance

- But a great starting point

When *everything* is an equity

- Everything becomes *mineable*

We are at the point where general *opinions* are actionable *positions* that we can leverage in sophisticated trading algorithms

The boundary between the Casino, Stock market, and Dining room conversations has entirely dissolved

- We no longer live our lives, we manage them as a collection of volatile assets

So…what do we focus on?

Introduction to Index Funds

Index funds are an investment funds designed to track a **specific market index**

- Instead of trying to pick individual winners an index fund holds all(or most) of the components in the underlying index
- They offer broad diversification
  - Reduces the risk that comes from relying on an individual stock performance
- Ex: The **S&P 500**
  - This is an aggregation of 500 individual stocks
    - Our initial focus is going to be on the S&P 500 as technical indicators usually work better on indexes compared to individual stocks
    - Index funds provide a baseline understanding of market behavior before diving into more active, high-frequency, or algorithmic strategies
    - They help frame why traditional market exposure differs so strongly from speculative assets or short-term trading

In the modern financialization of the US, performance of index funds **matter**

- Their performance dominates financial news and are viewed as broad economy markers

Correlation

Slide: CorrelationDo assets move together?Mining Context: If you go long on Google and long on Microsoft, are you actually diversified? (Likely not, correlation is high).

Technical Indicators

Beta (Relative Volatility)

Define Beta: A measure of how much a stock moves compared to the Index.Example: Beta of 1.5 means if the S&P 500 goes up 1%, this stock usually goes up 1.5%.

Moving Averages (Smoothing the Noise)

Momentum Oscillators (RSI)

Volatility Mining (Bollinger Bands)

Add a bold statement for a key point in your presentation

A word of caution before moving on

It's at this point in the semester where I feel it is my **civic duty** to remind us all:

- Do **not** engage in what this course is going to simulate with real income
  - There are **plenty** of ways to simulate trades to test models and trading systems
  - If you are testing methodologies with real capital, you have already lost.
  - We **are** going to lose money, a **lot** of it, over this course. That is **okay.**
    - We are going to learn how to do this in a safe, managed way…while learning a new technology
- Investing should only be done with **expendable** income
  - Only invest what you can afford to lose
  - Consider an invested asset *gone*

Stocks are a **volatile** asset

- They are heavily affected by human factors
  - Sentiment, performance, natural disasters(to an extent), regulatory bodies

The following tradable assets are **incredibly volatile assets**

- Some with little regulation, some with zero regulation, some with outright insider trading, rug pulls